

Pascal's Triangle and a Digital View of the Binomial Theorem

From Binomial Coefficients to Sierpiński Patterns

2015.12.05, UNH Math Department



Small rules $\rightarrow$ Large patterns

Small Local Rules Create Large Global Structures

Binomial Theorem → Pascal's Triangle → Modulo-2 Patterns → Sierpiński Structures



- Pascal's Triangle is not merely a table of coefficients.
- It is a recursive structure-generation mechanism.
- The parity of binomial coefficients reveals hidden digital patterns.

Reading Coefficients Through Combinatorics

The coefficient of each term is interpreted as the number of ways to choose k items from n .

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

$$x^4$$

1

$$4x^3y$$

4

$$6x^2y^2$$

6

$$4xy^3$$

4

$$y^4$$

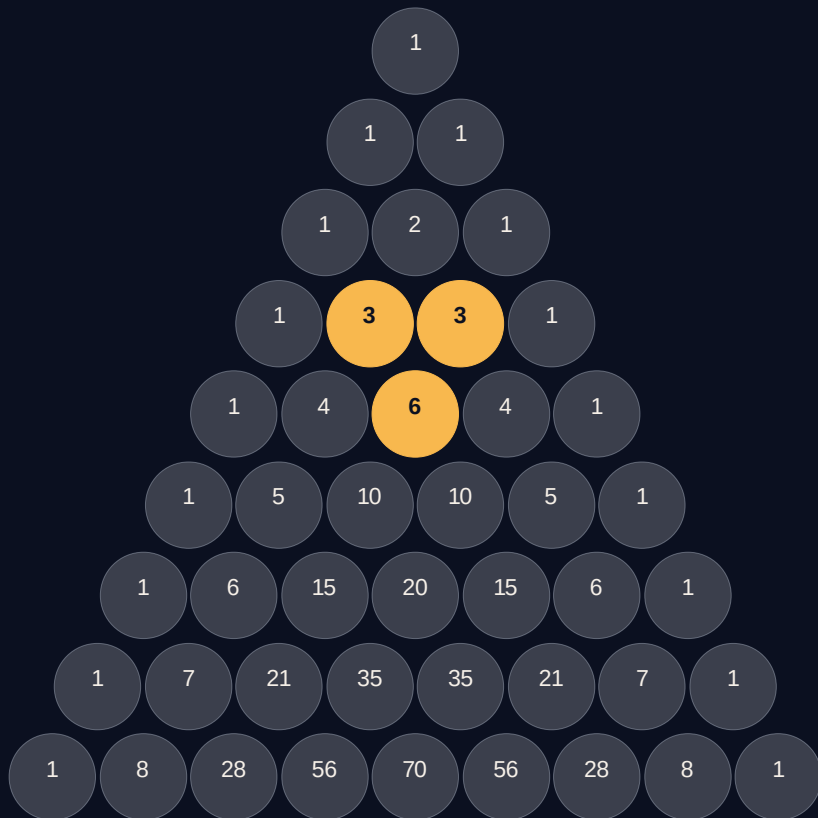
1

$$(x + y)^4$$

Coefficients in the Expansion = Pascal row 4

PASCAL'S TRIANGLE: A Map of Recursively Generated Coefficients

The number below is formed by adding the two numbers above it.



$$C(n,k) = C(n-1,k-1) + C(n-1,k)$$

예: $C(4,2) = C(3,1) + C(3,2) = 3 + 3 = 6$

- The computational rule is highly localized.
- However, the entire triangle exhibits symmetry and combinatorial structure.
- When this structure is viewed mod 2, an entirely different pattern emerges.

COMBINATORIAL INTERPRETATION: “Choosing k Objects from n”

Binomial coefficients are both coefficients in an algebraic formula and the number of ways to choose.



Choosing 2 Objects from 5

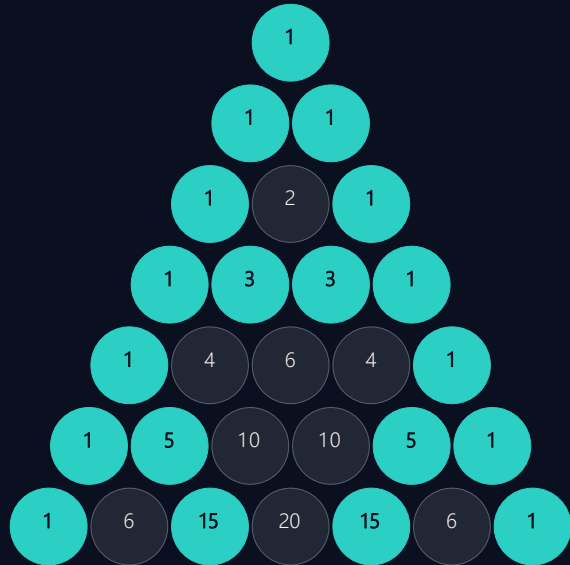
$$C(5,2) = 10$$

$$(x + y)^5 = x^5 + 5x^4y + 10x^3y^2 + \dots$$

The number of choices is the coefficient. This is the combinatorial meaning of the binomial theorem.

Hidden patterns emerge when only the odd numbers remain

When viewed modulo 2, Pascal's triangle forms a digital structure of 1s and 0s.



odd = 1

even = 0

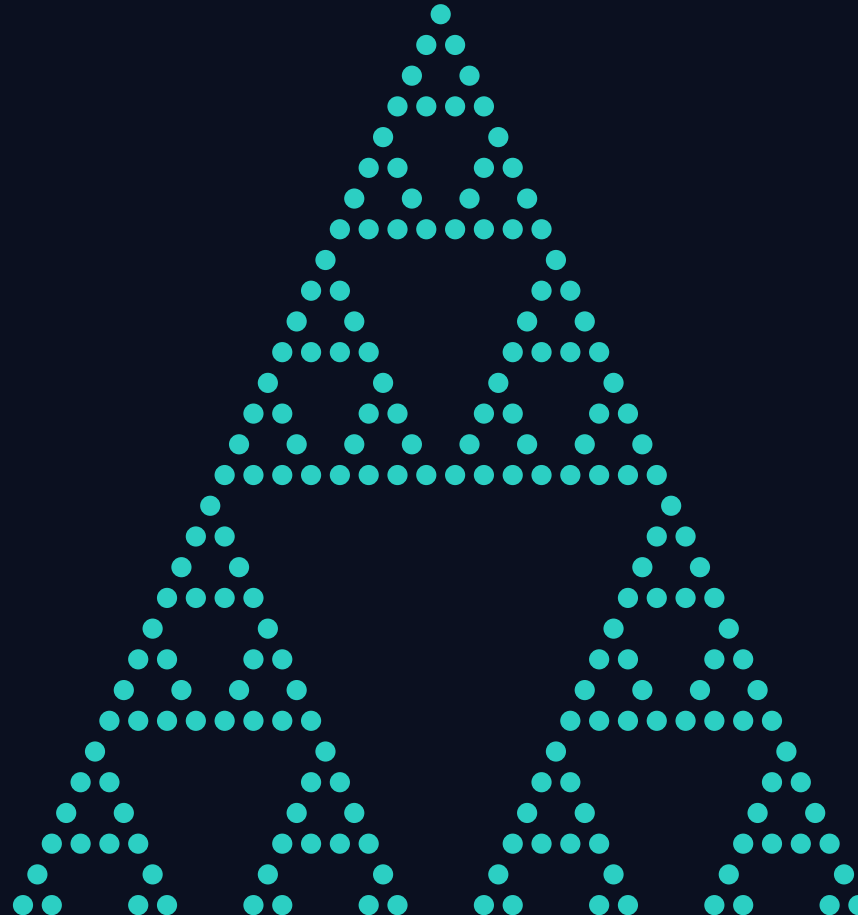
Discarding the magnitude of the coefficients and focusing solely on the parity makes the patterns much clearer.

SIERPIŃSKI PATTERNS IN PASCAL'S TRIANGLE

Fractal Geometry Hidden in Pascal's Triangle

Recursiveness

As the rows get larger, the same triangle pattern repeats in a smaller size.



Self-similarity

The whole and the parts have the same structure. The calculation table is a pattern generator itself.

Why does this pattern emerge?

The parity of binomial coefficients is controlled by binary digits.

$$C(n,k) \equiv \prod C(n_i, k_i) \pmod{2}$$

It can be explained by the mod 2 version of Lucas's theorem: you just need to check the validity of the choice for each binary bit.

- If a 1-bit of k is located at a 0-bit position of n , then $C(n,k)$ is even.
- In other words, k yields an odd coefficient only when it is chosen strictly within the bit-mask of n .
- This is the digital principle behind the Sierpiński pattern.

Ex: $n = 13 = 1101_2$



Odd Index k : 0, 1, 4, 5, 8, 9, 12, 13

Thus, odd coefficients only appear for k values that are formed by combining the 1-bit positions of n .

THREE INTERPRETATIONS

The same formula can be read as an algebraic, combinatorial, and digital pattern.

Algebraic

Expansion, Coefficients,
Formula

Combinatorial

Counting, Selection,
Combinations

Digital

Binary(Modulo 2),
Pattern Structure,
Fractals

When a single formula can have multiple interpretations, mathematics becomes a language for seeing structures rather than just calculation.

n=7



From Mathematical Structure to Workflow Systems

This seminar topic can be expanded into future workflow and IT Management interests.



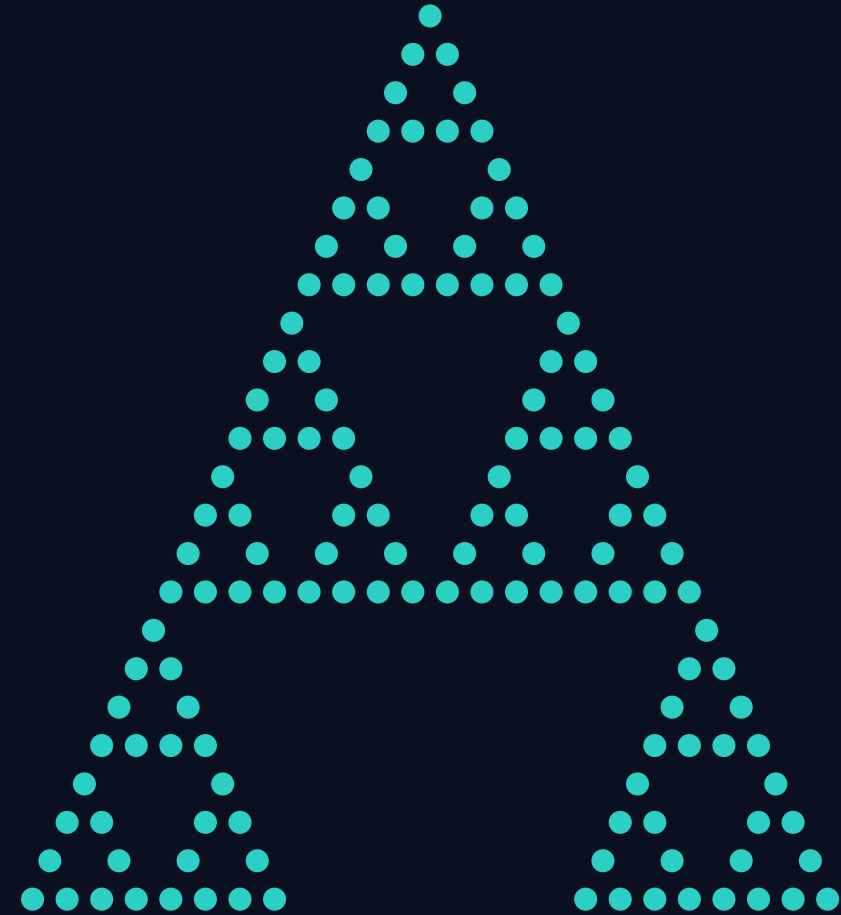
- Pascal's Triangle: Simple addition rules create the entire pattern.
- Organizational Workflow: Micro-rules of roles, approvals, and handoffs create the entire operational structure.
- AI Systems: Structuring documents, logs, and signals leads to intelligent decision-making.

Small Rules Create Large Patterns

Pascal's Triangle is not only a table of coefficients.
It is a compact model of recurrence, structure, and digital
pattern formation.

Q & A

Digital Interpretation of Pascal's Triangle and the Binomial
Theorem



References

Reference materials and keywords for restructuring your presentation

- Hieu D. Nguyen, “A Digital Binomial Theorem,” arXiv:1412.3181, 2014.
- Toufik Mansour & Hieu D. Nguyen, “A Digital Binomial Theorem for Sheffer Sequences,” arXiv:1510.08529, 2015.
- Lucas Theorem; Pascal’s Triangle modulo 2; Sierpiński triangle.
- Key Concepts: Binomial Theorem, Binomial Coefficients, Pascal Recurrence, Combinatorial Proof.

Core Keywords: Binomial coefficients · Pascal Recurrence · Modulo 2 · Digital Patterns